Process Characterization Plan

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCP-005

Process Development / Manufacturing Science & Technology (MSAT)

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Author	Process Development / MSAT	—	—	—
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Abbreviations

AEX, anion exchange chromatography. ALCOA+, attributable, legible, contemporaneous, original, accurate and complete. AMV, analytical method validation. BLA, Biologics License Application. CCD, central composite design. CEX, cation exchange chromatography. CQA, critical quality attribute. Cpk, one-sided process capability index. CV, column volume. DNA, deoxyribonucleic acid. DoE, design of experiments. ELISA, enzyme-linked immunosorbent assay. Fc, crystallizable fragment of the antibody. HCP, host cell protein. ICH, International Council for Harmonisation. IgG1, immunoglobulin G subclass 1. MSAT, Manufacturing Science and Technology. NOR, normal operating range. OLS, ordinary least squares. PAR, proven acceptable range. PD, Process Development. PPQ, process performance qualification. QA, Quality Assurance. qPCR, quantitative polymerase chain reaction. RSM, response surface methodology. SEC, size exclusion chromatography. SOP, standard operating procedure. UF/DF, ultrafiltration and diafiltration.

1 Purpose and scope

This plan describes the process characterization studies that will be performed on the Protein A capture step of the A-Mab drug substance process. Protein A chromatography is the first chromatographic unit operation of the purification train. It binds the antibody from the clarified harvest, removes the bulk of the host cell protein and DNA in the flow-through and the wash, and delivers a low-pH eluate to the viral inactivation step. The studies will be executed on a qualified scale-down model of the commercial column and will be reported in PCR-005.

The plan covers the 6 process parameters of the capture step listed in §6.1 and the 3 responses measured on the eluate pool. Of those parameters, 4 will be studied in a multivariate design and 2 will be assessed one at a time. The commercial process is operated at a production bioreactor scale of 15,000 L, and the capture step processes the clarified harvest of one such batch. The work is a Stage 1 process design activity under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and follows the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012).

Three subjects lie outside this plan. The validation of the analytical methods is complete and is covered by the method validation reports listed in §3. The resin life cycle, which includes the cycle number at which a column is retired and the sanitization regime applied between cycles, is governed by SOP-2008 and is not varied in these studies. Process performance qualification at the receiving site is Stage 2 work and is planned in PTP-001. No viral clearance claim is made for the capture step, and no viral clearance study is planned here.

2 Objectives

The studies planned here have five objectives.

1. Quantify the effect of each characterized parameter on pool host cell protein, on step yield and on leached Protein A, over the ranges given in §6.1.
2. Define the region of the characterized space within which the quality attributes this step controls are predicted to remain within their acceptance criteria.
3. Establish a proven acceptable range for each parameter and each quality attribute the step controls, both with the other parameters at their set-points and with the other parameters varying within their normal operating ranges.
4. Provide the demonstrated effects and the operating region from which each parameter will be classified under SOP-4001.
5. Justify the parameter ranges carried into Stage 2 and the contribution this step makes to the control strategy of the drug substance process.

The classification of the parameters is an outcome of the studies. It is assigned in PCR-005 and is not stated in this plan.

3 Related documents

This plan sits under the transfer plan and the master plan, and takes the scope of its studies from the pre-characterization risk assessment. The related documents of the package are given in Table 3.1.

Table 3.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-005	Process Characterization Report (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and validated methods that govern the work are given in Table 3.2. SOP-2007 governs the operation of the step, SOP-2008 the packing and life cycle of the column, SOP-1001 the qualification of the scale-down model, SOP-1002 the design and analysis of the designed experiments, and SOP-4001 the classification of the parameters once the results are available.

Table 3.2: Controlled procedures and validated analytical methods for the capture step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP

Reference	Title	Type
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Protein A capture is a bind and elute affinity step. Staphylococcal Protein A immobilized on the resin binds the Fc region of the antibody at neutral pH through hydrophobic and hydrogen bonding contacts at the interface between the second and third constant domains of the heavy chain. Host cell protein, DNA and most other components of the clarified harvest do not bind and leave in the flow-through or the wash. Lowering the pH protonates histidine residues at the interface between the Fc and the ligand, weakens the interaction and releases the antibody into a small volume of low-pH eluate. The step therefore concentrates the product, removes the bulk of the process impurities, and introduces one impurity of its own, which is ligand that leaches from the resin.

The capture step is a platform operation. It has been used for the manufacture of three previous humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical and engineering manufacture, and the chemistry of the resin, the buffers and the cycle are unchanged for A-Mab (CMC Biotech Working Group 2009). The mechanisms below are therefore known before the study begins. Characterization confirms and bounds them for this molecule rather than discovering them.

Host cell protein reaches the eluate pool by two routes. Some host cell protein binds the resin backbone or the ligand weakly and elutes when the pH is lowered. Some associates with the antibody itself and is carried through as a complex. Both routes depend on how much antibody is bound and how completely it is released. As the protein load approaches the dynamic binding capacity of the resin, the mass transfer zone extends further down the bed and impurity that would have been washed out is carried into the eluate. A lower elution pH releases weakly bound host cell protein together with the product. The two effects reinforce each other, so an interaction between protein load and elution pH is expected.

DNA leaves the step by a different route from host cell protein. DNA is polyanionic and does not bind Protein A, so it is removed in the load flow-through and in the wash, and the small fraction that reaches the eluate is DNA associated with the antibody or with host cell protein complexes. Clearance is large and is expected to be insensitive to the operating parameters within the ranges studied. Residual DNA is therefore measured to confirm clearance rather than to fit a model.

Leached Protein A is formed by the step itself. Ligand leaves the resin by hydrolysis of the immobilization chemistry and by proteolysis of exposed ligand, and the leached fragments elute with the antibody because they still bind Fc. The amount depends on the coupling chemistry of the resin, on the cumulative cycle number and on the sanitization history, all of which are properties of the resin over its life rather than of a single run. Within the operating ranges of one run the level is therefore expected to be flat. It is controlled by the resin life-cycle limits of SOP-2008 and cleared further downstream by the cation and anion exchange steps.

Step yield is lost by two routes as well. Antibody breaks through during the load, and antibody is left on the column at the end of elution. Breakthrough rises as the protein load approaches the dynamic binding capacity and as the residence time falls, because binding at the ligand is limited by diffusion into the pores of the bead. Extending the pool collection recovers the tail of the elution peak.

A higher protein load and a lower elution pH both raise pool host cell protein, and a faster load flow rate and a shorter pool collection both lower step yield. Protein load is the mass of antibody applied per volume of resin. Raising it fills a larger fraction of the binding capacity of the bed, which extends the mass transfer zone toward the column outlet, carries host cell protein that the wash would otherwise have removed into the eluate, and lets antibody break through into the flow-through. Lowering the elution buffer pH weakens the interaction between the Fc and the ligand further and releases the antibody sooner. The histidine residues at that interface are already largely protonated across the elution range, so within it a lower pH acts by protonating the acidic side chains that form the remaining hydrogen bonds and salt bridges. The antibody then desorbs sooner and in a sharper peak, and host cell protein held weakly on the resin or on the antibody is released with it. A higher pH within the range releases less host cell protein with the product and gives a broader, later peak and a larger pool volume. Over the ranges studied, elution pH is expected to change pool host cell protein more than any other parameter. Load flow rate sets the residence time and hence how far into the pore the antibody diffuses before the fluid moves on, so a faster load broadens the mass transfer zone, lowers the dynamic binding capacity and costs yield at high load. End of pool collect decides how much of the tail of the elution peak is taken. Because the weakly held impurities elute early in the low-pH transition rather than in the tail, extending the collection is expected to raise yield and to lower the impurity concentration per mass of antibody.

Operating temperature and bed height are expected to change pool host cell protein and step yield less than the multivariate parameters do. A higher operating temperature raises the diffusion coefficient of the antibody, which carries it faster into the pores of the bead, and it strengthens the hydrophobic contacts at the interface with the ligand, which holds the antibody more tightly during elution. The two effects act on yield in opposite directions, and over the range of 15 to 30 °C both are small beside the effects of load and elution pH. The binding capacity specified by the resin manufacturer covers that range. A taller bed at a fixed linear velocity lengthens the residence time, raises the dynamic binding capacity and raises the pressure drop across the column. Over the range of 10 to 30 cm the linear velocity is adjusted at scale-up so that residence time is held constant, and bed height is then expected to leave pool host cell protein and step yield unchanged.

4.2 Quality attributes in scope

The capture step forms one drug substance quality attribute of its own, which is leached Protein A. Table 4.1 gives that attribute with its acceptance criterion and its criticality.

Table 4.1: Quality attribute set by the capture step.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0–5 ppm	L-M	16

The step also carries the principal clearance of two impurities formed upstream, and these are given in Table 4.2. Host cell protein is the attribute that drives the design, because the capture step removes most of the host cell protein in the clarified harvest and because every nanogram it leaves in the pool has to be removed by the cation and anion exchange steps that follow. Residual DNA is cleared by the same step without a parameter dependence that the ranges studied are expected to expose.

Table 4.2: Quality attributes cleared by the capture step.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

Criticality was assigned in RA-001 from the potential impact of the attribute on safety and efficacy combined with the uncertainty in that assessment, which is the Tool #1 score shown in the two tables. Attributes scored high or very high are called critical. The outcome is a continuum of criticality rather than a binary split (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b). Leached Protein A is scored low to moderate. It is the attribute this step forms, it is measured in every run of both designs, and its acceptance criterion is a drug substance criterion.

Two attributes are measured across the step without a claim being made for them. Aggregate is measured by size exclusion chromatography on the load and on the pool to confirm that capture does not change it, and no aggregate clearance is claimed for this step. Viral clearance can be demonstrated across a Protein A step, and no claim is made for it here. The clearance claim of the drug substance process rests on the low-pH inactivation, anion exchange and virus filtration steps, and it is consolidated in PCMR-001 (International Council for Harmonisation 2023a).

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of the capture step before this plan was written. The ranking used the risk-ranking-and-filtering method of the A-Mab case study, in which each parameter is scored on its potential impact on a quality attribute and separately on its potential to interact with other parameters, and the two scores are combined into a study-type decision (CMC Biotech Working Group 2009). Where no data or rationale supported an assessment, the parameter was ranked at the highest level.

The assessment placed the parameters in three groups. Parameters with a credible direct effect on a quality attribute the step controls, and a credible interaction, went into the multivariate design, and there are 4 of them. Parameters whose effect is expected to be small over the range they can take, and whose interactions are not expected to be material, were assigned to univariate assessment, and there are 2 of them. Parameters ranked below the threshold for a new study are held to platform specification under SOP-2007 and are not varied here, and their ranges rest on prior knowledge. Buffer identity and quality, resin type and column packing quality fall in this last group.

The ranges to be studied are wider than the routine control ranges. Every characterized range in §6.1 is at least 2.0 times the width of the normal operating range of the same parameter. Two reasons support the expansion. A range wider than the control range bounds the knowledge space, so that an excursion during commercial manufacture can be assessed against data rather than against judgement. A wider range also lets the response surface resolve curvature, which a design run only across the control range would confound with noise.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be executed on a scale-down model of the commercial capture column, qualified under SOP-1001. Bed height, linear velocity and therefore residence time, protein load per volume of resin, buffer composition, and the wash and elution volumes normalized to column volumes are set at the commercial values. Column packing will be verified by plate count and peak asymmetry before use under SOP-2008, and the resin will be taken from the same manufacturing family as the commercial resin and used within the cycle number qualified under that procedure.

Qualification will compare the model against at-scale data from the engineering and clinical campaigns. The comparison covers step yield, pool host cell protein, leached Protein A, residual DNA and column efficiency. The model will be accepted as qualified where the small-scale means show no statistically significant difference from the at-scale values at $p < 0.05$ and the small-scale results fall inside the range of the at-scale data. Where a difference is found, its cause should be investigated and the model modified before the characterization studies begin (U.S. Food and Drug Administration 2011).

Replicated centre points carry the qualification into the studies themselves. The screening design carries 3 centre points and the response-surface design carries 4, all run at the step set-points. Their spread is the pure error of the system, and it is the term against which lack of fit is tested in §5.5. It also shows whether the model drifted over the execution of a design.

Two bounds apply to everything the model supports. The model predicts mean performance at commercial scale, and it does not reproduce the hydrodynamic variation of a commercial column or the variation between commercial harvest lots. The ranges established from it will be confirmed at scale during Stage 2.

5.2 Operation

Each run will follow the commercial cycle described in SOP-2007. The column is equilibrated, loaded with clarified harvest to the target protein load, washed, and eluted with the low-pH elution buffer. Pool collection starts at a fixed absorbance threshold on the ascending edge of the elution peak and ends at the column volume set by the end of pool collect parameter. The column is then stripped, sanitized and stored under SOP-2008.

The clarified harvest for a design will be taken from a single pooled lot, so that variation in the feed does not confound the factor effects. The lot will be characterized for titer, host cell protein, DNA and aggregate before the design is executed, and those values will be reported in PCR-005 as the input condition of the study. Buffers, resin and harvest are controlled to platform specification. They are identity and quality controlled rather than characterized, and they are held constant within a design.

5.3 Analytical methods

The validated methods that will generate the study data are listed in Table 5.1. Host cell protein in the eluate pool will be measured by the enzyme-linked immunosorbent assay of AMV-3012 and reported per milligram of antibody. Leached Protein A will be measured by the enzyme-linked immunosorbent assay of AMV-3016 and reported in parts per million of antibody. Residual DNA will be measured by quantitative PCR under AMV-3014. Aggregate will be measured by size exclusion chromatography under AMV-3011 on the load and on the pool. Step yield will be calculated from the antibody mass in the eluate pool and the mass loaded, each determined by ultraviolet absorbance.

Table 5.1: Validated analytical methods and the responses they measure.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

The precision of an assay sets a floor on the effect a study can resolve. Leached Protein A is the response to watch, because it is judged against a criterion of 5 ppm and the mechanism in §4.1 gives no reason to expect a parameter effect on it. The precision of each assay at the level measured will be reported with the results, and an effect smaller than the reported precision of its assay will not be claimed as an effect. Method performance detail is held in the method validation reports and is not repeated in PCR-005.

5.4 Sampling plan

Every run of both designs will be sampled at four points. The clarified harvest load will be sampled for titer, host cell protein, DNA and aggregate. The flow-through will be sampled to quantify antibody breakthrough. The wash will be sampled for host cell protein. The eluate pool will be sampled for antibody concentration, host cell protein, leached Protein A, residual DNA and aggregate. Sample volumes, containers and storage conditions are given in SOP-2007, and retains will be held frozen so that a run may be re-tested if an assay result is questioned.

The 47 runs of the two designs will be executed in randomized order within each design, and the centre points will be distributed through the execution sequence rather than run consecutively. A drift in the system then appears as a trend across the centre points in run order, and the pure error estimate stays free of it.

5.5 Statistical methods

All analyses will be performed under SOP-1002. Each factor is coded over its characterized range, with the low edge at −1, the high edge at +1 and the centre at 0. The centre of each characterized range in §6.1 is the set-point of the parameter, so the centre points of both designs are runs at the step set-points. Coding puts the factors on a common scale.

The screening data will be analysed by ordinary least squares with all main effects and all two-factor interactions in the model. The effect of a term is twice its coded coefficient, which is the change in the response across the full studied range of that factor. A term will be reported as significant at $p < 0.05$. A factor with no significant term over the range studied will be reported as having no detectable effect on that response, and the absence will be stated together with the range over which it was tested.

The response-surface data will be fitted with the full quadratic model in the 4 multivariate factors. Model adequacy will be judged on four measures. The model F test states whether the model explains more than noise. The coefficient of determination, its adjusted form and the predicted form computed from the prediction error sum of squares state how much of the variation is explained and how much of it survives when each run is left out in turn. The analysis of variance partition of residual variance into lack of fit and pure error states whether the model form is adequate for the data, using the centre points of §5.4 as the pure error term. Diagnostic plots of residuals against predicted values, of residual quantiles against normal quantiles, and of actual against predicted values will be reported for each modelled response.

The screening design identifies which factors and interactions matter. The response-surface model is the predictive model, and it is the model from which the operating region, the proven acceptable ranges and the capability estimates in §7 and §8 are derived. Effect magnitudes quoted from the screening analysis are superseded by the response-surface estimates wherever the two disagree.

Capability at commercial scale will be estimated by Monte-Carlo simulation of 2,000 batches, with each parameter drawn within its normal operating range and the fitted response-surface model used to predict the attribute. Capability will be reported as a one-sided Cpk against the criterion of the attribute, because every quality attribute this step controls has a ceiling and no floor.

6 Study design

6.1 Factors, ranges and study type

The parameters to be studied, their set-points, the ranges to be studied and the routine control ranges are given in Table 6.1. The study type in the last column is the outcome of the risk-based prioritization described in §4.3.

Table 6.1: Parameters, ranges and study type for the capture step.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	30	10–50	20–40	multivariate
Elution buffer pH	pH	3.55	3.2–3.9	3.4–3.7	multivariate
Load flow rate	cm/hr	200	100–300	150–250	multivariate
End of pool collect	CV	2.6	2–3.2	2.3–2.9	multivariate
Operating temperature	°C	22	15–30	18–26	univariate
Bed height	cm	20	10–30	15–25	univariate

Protein load will be studied over 10 to 50 g/L resin against a set-point of 30 g/L resin, which takes the high edge of the design toward the dynamic binding capacity of the resin where breakthrough is expected. Elution buffer pH will be studied over pH 3.2 to 3.9 against a set-point of pH 3.55. Load flow rate will be studied over 100 to 300 cm/hr and end of pool collect over 2 to 3.2 CV, both of which cover more than the excursion a commercial chromatography skid can produce. The coded letters used for the factors in the design matrices and in the effect tables are given in Table 6.2.

Table 6.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

6.2 Screening design

The screening study is a two-level full factorial in the 4 multivariate factors with 3 centre points, which is 16 factorial runs and 19 runs in total. A full factorial was selected rather than a fractional design because four factors are few enough to afford it, and because it estimates every main effect and every two-factor interaction without aliasing. The interaction between protein load and elution pH described in §4.1 is the reason the interactions matter here, and a resolution IV design would have aliased it with another two-factor interaction. The planned matrix is given in Appendix A.

A two-level design cannot estimate curvature. The centre points test for curvature as a single degree of freedom, and they cannot say which factor carries it. Where curvature is present, the response-surface design resolves it. For this reason the screening analysis will be reported as the identification of effects and their directions, and not as a predictive model.

6.3 Response-surface design

The response-surface study is a face-centred central composite design in the same 4 factors, with 16 factorial runs, 8 axial runs and 4 centre points, which is 28 runs in total. The axial points sit on the faces of the cube rather than outside it, so no run exceeds the characterized range of its factor and every setting stays inside a region the equipment can hold. The design supports the full quadratic model, which adds the squared terms that describe curvature to the main effects and interactions already estimated by the screening design. The planned matrix is given in Appendix B.

The response-surface design covers all 4 multivariate factors and is therefore fixed before the screening data are seen. Prior knowledge is sufficient here to justify carrying every multivariate factor forward, and executing both designs on the same factor set means the two datasets can be compared directly. Where the screening study shows a factor to have no effect on any response, that finding will be reported, and the factor will still appear in the response-surface model so that the model covers the whole characterized region.

6.4 Univariate assessment

Operating temperature and bed height will be assessed one factor at a time. Each will be run at the low edge, the centre and the high edge of its characterized range, with the other parameters held at their set-points and with the same responses measured as in the multivariate designs. Temperature will be assessed over 15 to 30 °C and bed height over 10 to 30 cm.

These two parameters were kept out of the multivariate design for the mechanistic reasons given in §4.1. Their expected effects are small beside those of load and elution pH over the ranges they can take, and neither is expected to interact with the parameters that drive host cell protein. A univariate study cannot detect an interaction, and this limitation is accepted here because the risk assessment found no credible interaction to detect. Should the univariate data show an effect comparable with the multivariate effects, the parameter will be added to a supplementary multivariate study before the operating region is declared.

7 Acceptance and decision criteria

This section states the criteria the study must meet. It states no outcome, and every criterion below is fixed before the data are generated.

The scale-down model must meet the qualification criteria of §5.1 before any characterization run is executed. A characterization run performed on an unqualified model may not be used to support an operating range.

A response-surface model will be judged adequate for a response when its model F test is significant at $p < 0.05$, when its lack of fit is not significant against pure error at $p < 0.05$, and when its adjusted and predicted coefficients of determination are reported and consistent with each other. A large gap between the adjusted and predicted forms indicates a model that fits the executed runs better than it predicts a new one, and where such a gap is found the model will be re-examined before it is used. Where lack of fit is significant, the model will not be used to define the operating region until the misfit is explained, and the design may be augmented under SOP-1002.

The acceptance criteria against which the responses will be judged are given in Table 7.1. Pool host cell protein is judged against an in-process limit rather than against the drug substance criterion, because the cation and anion exchange steps still clear host cell protein after the capture step and the capture pool is an intermediate. The in-process limit is the drug substance criterion of 100 ng/mg carried back through the clearance the downstream steps deliver in the nominal train, divided by a safety factor of 4.5. Without the safety factor the limit would sit at the concentration above which the downstream steps can no longer bring the batch to the drug substance criterion, and the drug substance would then depend on every downstream step delivering its nominal clearance exactly. Leached Protein A is judged against the drug substance criterion directly, because it is the attribute this step forms and the criterion is nearly binding at the capture pool.

Table 7.1: Acceptance criteria for the measured responses.

Response	Acceptance criterion	Basis
Pool HCP (ng/mg)	≤ 10,724 ng/mg	In-process limit, back-calculated from the drug substance criterion of 100 ng/mg
Leached Protein A (ppm)	≤ 5 ppm	Drug substance criterion
Step yield	None	Process performance attribute, reported and not judged against a quality criterion

Step yield carries no quality acceptance criterion. It will be reported for every run and used in the discussion of the operating region. A setting that meets both impurity criteria at an unacceptable yield is not a setting the process would be operated at.

The operating region will be declared as the part of the characterized space over which every quality attribute the step controls is predicted by its response-surface model to meet the criterion in Table 7.1. The study passes when the model is adequate for each modelled response, when the normal operating range of every parameter lies inside that region, and when the proven acceptable ranges of §8 contain the normal operating range. A parameter whose normal operating range is not contained will have its range narrowed in PCR-005, and the narrowed range will be carried into the control strategy.

Capability will be estimated and reported for each quality attribute the step controls, as described in §5.5. No numerical capability threshold is set in this plan, because no such threshold is established for these attributes in this process. The estimate will be read together with the margin between the predicted level and its criterion, and both will be reported.

Any departure from this plan will be recorded as a deviation at the time it occurs, with the runs it affects, an assessment of its impact on the conclusions, and its disposition. Deviations will be reported in PCR-005 whether or not they affect the outcome. Where a deviation invalidates a design for the definition of the operating region, the affected design will be re-executed and the original dataset retained and referenced.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of a single parameter over which the quality attribute it affects is shown by data to remain acceptable (U.S. Food and Drug Administration 2011). It is determined per attribute and per parameter, so one parameter may carry several proven acceptable ranges, and the narrowest of them is the one that binds. The analysis described here will be run for every combination of a quality attribute the step controls and a multivariate parameter, and its results will be reported in PCR-005.

The acceptance basis is the criterion in Table 7.1. Pool host cell protein will be judged against the in-process limit and leached Protein A against the drug substance criterion, and the criterion used will be stated in the reported table beside each range so that the reader does not have to infer which yardstick was applied. Step yield is not a quality attribute and therefore has no proven acceptable range.

Two analyses will be run for each combination. The first holds the other parameters at their set-points, which for every multivariate parameter of this step is the centre of its characterized range, and scans the parameter across its characterized range using the mean prediction of the fitted response-surface model. The second propagates the variation of the other parameters. It scans the parameter across 81 settings, and at each setting it draws 2,000 batches with the other parameters sampled within their normal operating ranges and with the residual variation of the fitted model added. The range from the second analysis is expected to be the narrower of the two, because it asks the attribute to stay acceptable under the variation the process actually carries rather than at one nominal setting of the other parameters.

The decision rule is the same for both analyses. The proven acceptable range is the contiguous interval around the centre over which the response meets its criterion, evaluated on the mean prediction for the first analysis and on the 95 percent predictive interval for the second. An interval that meets the criterion but is separated from the centre by settings that do not is not part of the range. A range is bounded by the characterized range of the parameter and will never be reported beyond it, since the model has no data outside the region it was fitted on.

Each quality attribute the step controls will be reported with a plot of the response against the parameter that has the largest main effect on it in the response-surface model, with the acceptable region shaded green and the set-point and the normal operating range marked, so that the margin between the routine operating range and the edge of acceptance can be read directly.

A proven acceptable range is a univariate statement and does not replace the operating region of §7. Two parameters may each sit inside their own proven acceptable range while their combination sits outside the multivariate region. Where the two disagree, the multivariate region takes precedence, and the proven acceptable ranges will be reported as supporting information for excursion assessment.

9 Data management and integrity

All study data will be handled under the ALCOA+ principles. Chromatography data will be acquired in the validated chromatography data system, analytical results will be recorded in the laboratory information management system, and run conditions and observations will be recorded in the electronic laboratory notebook. Each system carries an audit trail, and records are attributable to the analyst who made them and contemporaneous with the work.

Raw data will be retained in the acquiring system and will not be transcribed by hand into the analysis. Datasets used for the statistical analysis will be exported under a controlled procedure, and each export will be reconciled against its source records by a second person before the analysis is run. The statistical analysis, including the scripts and their input datasets, will be retained so that the analysis in PCR-005 can be reproduced from the raw data.

Any change to a recorded value will be made as a tracked correction with its reason, and the original value will remain visible. Data that are excluded from an analysis will be reported with the reason for the exclusion, and no result may be excluded because of its value alone.

10 Roles and responsibilities

The functions responsible for the work are given in Table 10.1. The author of this plan is accountable for its execution as written and for raising a deviation where execution departs from it.

Table 10.1: Functions and their responsibilities for the planned studies.

Function	Responsibility
Process Development / MSAT	Owns the plan, qualifies the scale-down model, executes the runs, interprets the results and writes PCR-005
Statistics	Specifies the designs, performs the model fitting and the capability analysis, and reviews the statistical conclusions
Analytical Development	Performs the validated assays, reports method performance at the levels measured, and investigates questioned results
Manufacturing Science	Reviews the ranges for consistency with the commercial equipment and the batch record, and receives the control strategy input
Quality Assurance	Approves this plan and PCR-005, and reviews deviations and their dispositions

11 Deliverables and schedule

The deliverable of this plan is PCR-005, the process characterization report for the capture step. The report will present the qualification of the scale-down model, the executed designs, the fitted models with their adequacy, the operating region, the proven acceptable ranges, the capability estimates, the classification of each parameter and the contribution of the step to the control strategy. The conclusions of PCR-005 roll up into PCMR-001.

The work will be executed in five phases and in the order given. The scale-down model is qualified first. The screening design is executed and analysed second. The response-surface design is executed third, and it will not begin until the screening data have been reviewed and the model qualification remains valid. The univariate assessments are executed fourth and may run in parallel with the response-surface design, since they share no equipment constraint with it. The analysis and the report are completed last.

No calendar dates are fixed in this plan. The sequence above is the binding commitment, and the schedule is held in the project plan under PTP-001.

12 Risks and assumptions

The plan rests on four assumptions. The scale-down model is assumed to represent the commercial column for the attributes measured, and that assumption is tested by the qualification of §5.1 rather than taken on trust. The clarified harvest used in the studies is assumed to be representative of commercial harvest, which is the reason the input attributes of the feed lot are characterized and reported. The platform knowledge summarized in §4.1 is assumed to transfer to A-Mab, which the case for prior knowledge in that section justifies by mechanism. The resin is assumed to behave within its qualified cycle window, which SOP-2008 controls.

Four risks are recognized. A feed lot that is not representative would bias every run of a design, and the mitigation is the characterization of the feed lot before the design is executed. A model that fails its qualification criteria would delay the studies, and the mitigation is to qualify the model before the designs are scheduled. An assay whose precision at the measured level is poor relative to the effects would limit what the study can resolve, which is why assay precision is reported with the results and why no effect smaller than that precision will be claimed. A significant lack of fit in a response-surface model would prevent the operating region from being declared for that response, and the mitigation is the augmentation route of §7.

One risk is accepted without mitigation. The operating region and the proven acceptable ranges will be established on a scale-down model, and they are not confirmed at commercial scale by this study. Confirmation is Stage 2 work under PTP-001, and the claims made in PCR-005 will be bounded accordingly.

13 Approvals

This plan requires the approval of the functions listed in Table 1 before any characterization run is executed. Approval signifies that the scope, the designs, the acceptance criteria and the decision rules are agreed as written, and that the results obtained under them will be accepted as the basis for the operating ranges of the capture step. A change to this plan after approval will be made by revision of the document, and a departure discovered during execution will be handled as a deviation under §7.

References

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Appendix A — Planned screening design matrix

The coded settings of the planned screening design are given in Table 13.1. The factor letters are those of Table 6.2, the level −1 is the low edge of the characterized range, +1 the high edge and 0 the set-point. Response columns are absent because no run has been executed.

Table 13.1: Planned screening design matrix, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Appendix B — Planned response-surface design matrix

The coded settings of the planned response-surface design are given in Table 13.2. The axial runs carry one factor at ±1 with the others at 0, which places them on the faces of the cube described in §6.3.

Table 13.2: Planned response-surface design matrix, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0